Invisible waves: Summary

Answers

Use the listed words to complete the sentences.

convex	focal	frequency	microwaves	opaque
plane	rays	reflection	scattered	speed
straight	translucent	transparent	transverse	

1. Light is a form of energy. Light travels in ……straight…… lines. The lines that are used to show the path of light are called ……rays………
2. Light travels at the ……speed…… of 300 000 kilometres per second in a vacuum.
3. Rays or beams of light may be seen when light is ……scattered…… or reflected from particles in substances and redirected to the eye.
4. Light reflected from a mirror follows the law of ……reflection…… which states: the angle of incidence equals the angle of reflection. Flat mirrors are called ……plane…… mirrors.
5. Curved mirrors may be concave (curved inwards) or ……convex…… (curved outwards). Light reflected from concave and convex mirrors also follows the law of reflection. Parallel rays of light are reflected to a ……focal…… point in a concave mirror. Convex mirrors reflect light from a wide angle creating an image reduced in size that captures much of the surrounds.
6. Shiny surfaces reflect light but if a surface absorbs light it is said to be ……opaque………
7. If most of the light travels through a material, the material is said to be ……transparent…… Window glass is transparent.
8. Some materials allow some light to travel through but also scatter some of the light. Objects are not clearly visible when viewed through these materials. Frosty glass used in bathroom windows is an example. Such materials are said to be ……translucent……
9. Light is an example of an electromagnetic wave. These waves are ……transverse…… waves. The distance from crest to crest is the wavelength of a wave.
10. The electromagnetic spectrum is divided into regions from low frequency to high frequency: radio waves; ……microwaves……; infra-red waves; visible waves; ultra-violet waves; X-rays; gamma rays.
11. Analogue and digital television, wireless broadband and mobile phone calls can all be transmitted over long distances using high ……frequency…… radio waves, also called microwaves.